TROUBLESHOOTING GUIDE

VC

VIRTUAL CONTROL

VMWARE CLOUD FOUNDATION SOLUTIONS

VCF 9.0 Troubleshooting Guide

Comprehensive VCF 9 troubleshooting guide covering SDDC Manager, vCenter, NSX, vSAN, ESXi, and VCF Operations — based on Broadcom TechDocs and KB.

VCF 9 | TROUBLESHOOTING | MULTI-COMPONENT | BROADCOM KBS

VCF 9.0 VMware Cloud Foundation

VCF 9.0 Troubleshooting Guide

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> **About this reconstruction.** This document was reconstructed from a PDF that had no Markdown source. The body below is the verbatim extracted text — page headers, page numbers, and file URL artifacts have been stripped, but **fixed-width formatting may not be perfect** and **section headings are not yet promoted to Markdown H2/H3 syntax**. Treat this as an editable starting point — clean up specific sections as needed. The PDF and HTML companion files are regenerated from this Markdown source going forward.

```
      VMware Cloud Foundation 9.0
Comprehensive Troubleshooting Guide

                   Based on Official Broadcom Documentation
                        For Broadcom Interview Preparation

Document Purpose:    VCF 9 Administrator Interview Preparation
Source:              Broadcom TechDocs and Knowledge Base
Components:          SDDC Manager, vCenter, NSX, vSAN, ESXi, VCF Operations
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Official Documentation Sources:
       - techdocs.broadcom.com/us/en/vmware-cis/vcf/vcf-9-0-and-later/9-0.html
       - knowledge.broadcom.com (VCF Knowledge Base)
       - Broadcom Support Portal

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1. VCF 9.0 Architecture Overview

VMware Cloud Foundation 9.0 integrates four core components into a unified SDDC platform:

Component	Purpose	Key Services
vSphere (ESXi 9.0)	Compute virtualization	hostd, vpxa, vmkernel
vSAN 9.0	Software-defined storage	vSAN daemon, CMMDS, CLOM
NSX 9.0	Software-defined networking	nsx-proxy, DFW, Edge
SDDC Manager	Lifecycle management	LCM, Domain Manager
VCF Operations	Monitoring (Mandatory)	Diagnostics, Alerts

VCF 9.0 Upgrade Order (Critical):

    SDDC Manager > vCenter > NSX > ESXi Hosts > vSAN > VCF Operations

IMPORTANT: SDDC Manager must be upgraded first as it enables features for subsequent component upgrades.

2. SDDC Manager Troubleshooting

SDDC Manager is the automation and lifecycle management engine for VCF. In VCF 9.0, the UI is being deprecated and moving to VCF Operations Fleet Management.

2.1 Services and Health Checks

Check All SDDC Manager Services:

```
systemctl status vcf-services
```

Or use the SoS utility:

```
/opt/vmware/sddc-support/sos --services-health --domain-name ALL
```

Key SDDC Manager Services:

Service	Function	Restart Command
commonsvcs	Common services	systemctl restart commonsvcs
lcm	Lifecycle Manager	systemctl restart lcm
domainmanager	Host commissioning	systemctl restart domainmanager
operationsmanager	Operations tasks	systemctl restart operationsmanager

Restart All Services:

```
systemctl restart vcf-services
```

```
2.2 Log File Locations           Location
                                 /var/log/vmware/vcf/
             Log Type            /var/log/vmware/vcf/domainmanager/domainmanager.log
             Primary VCF Logs    /var/log/vmware/vcf/lcm/lcm.log
             Domain Manager      /var/log/vmware/vcf/operationsmanager/
             LCM (Lifecycle)     /var/log/vmware/vcf/vdt/
             Operations Manager  /var/log/vmware/vcf/sddc-support/
             VDT Tool Logs
             SoS Bundles
```

2.3 VCF Diagnostic Tool (VDT)

The VDT is a read-only diagnostic tool that checks configuration and reports PASS/WARN/FAIL results.

Running VDT:

1. SSH to SDDC Manager as vcf user

2. Switch to root: su -
3. Run: python /opt/vmware/sddc-support/vdt.py
4. Enter SSO password when prompted

VDT Checks Performed:

- SDDC Basic Info and Inventory Status

- SDDC, vCenter, and NSX Certificates
- Disk Space Usage
- NFS share permissions and ownership
- Invalid password task status
- Manifest File and SDDC Depot Checks
- SDDC Locks

Source: knowledge.broadcom.com/external/article/314610

2.4 Common Issues and Resolutions

Issue	Symptom	Resolution
Locked Account	Cannot login	systemctl restart commonsvcs
Workflow Stuck	Task shows In Progress	Check logs, retry or cancel
vSAN HCL Outdated	Precheck fails	Update vSAN HCL database
NSX Upgrade Failed	Completed in NSX, failed in SDDC	Update version in DB, restart LCM
Password Sync	Credential ops fail	Use Remediate option

SDDC Manager Account Reset:

```
SSH to SDDC Manager, su to root, then: echo "" > /etc/security/opasswd
```


3. vCenter Server Troubleshooting

3.1 Service Management

Check All vCenter Services:

```
service-control --status --all
```

Stop All Services:

```
service-control --stop --all
```

Start All Services:

```
service-control --start --all
```

Key vCenter Services:	Description
Service	vCenter Server Core Service
vmware-vpxd	vCenter Database (PostgreSQL)
vmware-vpostgres	Reverse HTTP Proxy
vmware-rhttpproxy	Profile-Driven Storage
vmware-sps	vSAN Health Service
vmware-vsan-health	

3.2 Log Files and Diagnostics

Log	Location
vpxd (Core)	/var/log/vmware/vpxd/vpxd.log
vSphere Client UI	/var/log/vmware/vsphere-ui/logs/
Authentication	/var/log/vmware/sso/
Database	/var/log/vmware/vpostgres/

3.3 Common Issues

vpxd Service Failure After Upgrade:

1. Check service status: service-control --status --all
2. Review vpxd.log for specific errors
3. Check database: /opt/vmware/vpostgres/current/bin/psql -U postgres

If certificate errors appear, certificates may need regeneration via Certificate Manager.

4. NSX Troubleshooting

NSX provides software-defined networking including overlay networks, distributed firewall, and edge services.

4.1 Transport Node Issues

Common Transport Node Problems:

- Status shows Unknown or Not Available
- Tunnel status degraded
- Interface down causing degraded state
- Certificate validation failures

Troubleshooting Commands (ESXi Host):

```
# Check nsx-proxy status (primary in NSX 9.x)
/etc/init.d/nsx-proxy status

# Check NSX VIB installation
esxcli software vib list | grep nsx
```

If Unable to reach client error appears, verify DNS resolution to NSX Manager first.

4.2 Edge Node Issues

Edge HA Failover Issue:
Edge nodes may enter NSX Maintenance Mode during HA failover if there are datapath or heap memory issues.

Check edge cluster history: get edge-cluster history state

Redeploying Failed Edge Node:

1. Ensure NSX Manager connectivity to Edge is DOWN before replacement
2. NSX Manager UI: System > Fabric > Nodes > Edge Transport Nodes
3. Select failed node > Actions > Redeploy
4. Error 78006 indicates Edge is still connected - power off first

4.3 Distributed Firewall (DFW)

DFW Rule Processing Order (Highest to Lowest):

Emergency > Infrastructure > Environment > Application > Default

DFW Troubleshooting Tools:

- Traceflow: Trace packet flow through NSX networks
- Flow Monitoring: Visualize traffic patterns
- Security Dashboard: View rule effectiveness

Default DFW Action: ALLOW (if no rules match)

4.4 Connectivity Testing

Test TEP Connectivity with MTU:

```
vmkping -d -s 1572 [TEP_IP]
```

-d = Do not fragment, -s 1572 = Size for 1600 MTU test

Minimum MTU Requirement: 1600 bytes for NSX overlay (GENEVE)

5. vSAN Troubleshooting

5.1 Health Checks and Monitoring

vSAN Skyline Health:
Access via vSphere Client: Cluster > Monitor > vSAN > Skyline Health
Click Ask VMware button on any finding to open the relevant KB article.

CLI Health Check:

```
esxcli vsan health cluster list
```

Clusters with health score below 60 or critical alerts are flagged in VCF Operations.

5.2 vSAN Network Troubleshooting

Key vSAN Ports:

Port	Protocol	Purpose
2233	TCP	vSAN Cluster Service
23451	UDP	Heartbeat (sent every second)
12345	TCP	CMMDS updates (primary/backup)

Note: vSAN no longer supports multicast for network communication.

Essential Network Commands:

```
esxcli vsan network list # Show vSAN network config
esxcli network ip interface list # Display VMkernel interfaces
vmkping -d -s 8972 [IP] # Test jumbo frame connectivity
esxcli network ip neighbor list # Show ARP entries
```

5.3 Failure Handling

Non-Compliant Objects:
Most common cause: Insufficient hosts or disk space to satisfy the storage policy.

Host Not Responding:
vSAN waits for host recovery before rebuilding components elsewhere in cluster.

Recommended Slack Space: Maintain 30% free capacity for rebuilds and rebalancing.

6. ESXi Host Troubleshooting

6.1 Log Files

Log	Location	Purpose
vmkernel.log	/var/log/	VMkernel, drivers, hardware
hostd.log	/var/log/	Host daemon, VM operations
vpxa.log	/var/log/	vCenter agent communication
vobd.log	/var/log/	VMkernel observations
vsantraced.log	/var/log/	vSAN daemon trace

By default, ESXi logs are in-memory and lost on reboot. Configure logging to a datastore for persistence.

6.2 Service Management

Restart Management Agents:

```
services.sh restart
```

OR

```
/etc/init.d/hostd restart && /etc/init.d/vpxa restart
```

Key ESXi Services:
- hostd: Host daemon - VM management
- vpxa: vCenter agent - communication with vCenter
- nsx-proxy: NSX communication (primary in NSX 9.x)
- vsan-mgmt: vSAN management daemon

6.3 Common Commands

```
vim-cmd vmsvc/getallvms # List all VMs
esxcli network ip interface list # List VMkernel interfaces
esxcli system version get # Show ESXi version
esxcli software vib list # List installed VIBs
esxcli network ip dns server add --server=[IP] # Add DNS server
esxcli vsan health cluster list # Check vSAN health
```

```
7. VCF Operations Troubleshooting

VCF Operations is MANDATORY in VCF 9.0 and provides monitoring, diagnostics, and alerting across the
platform.

Key Capabilities:
       - Discover and remediate known issues across VCF components
       - Monitor operational state of private cloud infrastructure
       - Property-based and log-based findings detection
       - Log Assist: automatically upload support bundles to Broadcom

Troubleshooting Workbench:
Access: VCF Operations > Troubleshoot > Troubleshooting Workbench

     If actions are missing, verify collector is active and in green Collection Status.

8. Backup and Recovery Procedures

VCF components use file-based backups. Monitor SFTP server space utilization.

After SDDC Manager Restore:
       1. SSH to SDDC Manager
       2. Run: sudo /opt/vmware/sddc-support/sos --health-check
       3. Verify all tests show GREEN

After vCenter Restore:
Validate SDDC Manager state to ensure inventory is consistent with recovered VMs.

NSX Manager Node Recovery:
       1. Retrieve NSX Manager credentials from SDDC Manager inventory
       2. Detach failed node from cluster (get UUID first)
       3. Deploy new NSX Manager node
       4. Join new node to existing cluster
       5. Verify cluster health

9. SoS Utility Reference

The Supportability and Serviceability (SoS) utility is the primary diagnostic tool for VCF.
Location: /opt/vmware/sddc-support/sos

Health Check Options:

     --health-check --force # Comprehensive diagnostics
     --connectivity-health # Verify ESX, vCenter, NSX connectivity
     --certificate-health # Validate certificate expiration
     --ntp-health # Confirm time synchronization
     --services-health # Verify running services
     --services-health --domain-name ALL # Check all domains

Log Collection Options:

     --vcf-installer-logs # VCF Installer appliance logs
     --sddc-manager-logs # SDDC Manager diagnostics
     --collect-all-logs # All component logs
     --zip # Compress output
     --domain-name [name] # Collect logs for specific domain

SoS Bundle Location: /var/log/vmware/vcf/sddc-support/

     Timeout: SoS may timeout after 60 minutes on large domains. Use component-specific options.

10. Quick Reference Tables

Port Reference:

Port                 Protocol  Component            Purpose
22                   TCP       ESXi/SDDC/vCenter    SSH
443                  TCP       All                  HTTPS/UI/API
902                  TCP       ESXi                 NFC (Network File Copy)
2233                 TCP       vSAN                 Cluster Service
```

8000	TCP	ESXi	vMotion
1600+	UDP	NSX	Overlay (GENEVE) - min MTU

Command Reference:

Component	Command	Purpose
SDDC Manager	systemctl status vcf-services	Check VCF services
SDDC Manager	/opt/vmware/sddc-support/sos --health-checRk un health check	
vCenter	service-control --status --all	Check vCenter services
ESXi	services.sh restart	Restart mgmt agents
ESXi	vim-cmd vmsvc/getallvms	List all VMs
ESXi	vmkping -d -s 1572 [IP]	Test MTU connectivity
vSAN	esxcli vsan health cluster list	Check vSAN health
NSX	get edge-cluster history state	Edge HA status

Interview Preparation Tips:

- Always start troubleshooting by checking service status and reviewing logs
- Know the VCF upgrade order: SDDC Manager first, VCF Operations last
- DFW rule priority: Emergency > Infrastructure > Environment > Application > Default
- Minimum MTU for NSX overlay is 1600 bytes
- vSAN requires 30% slack space for healthy operations
- VCF Operations is now mandatory in VCF 9.0
- The SoS utility is your primary diagnostic tool - know the options
- VDT is a read-only diagnostic tool for quick health assessment

Official Documentation:
techdocs.broadcom.com/us/en/vmware-cis/vcf/vcf-9-0-and-later/9-0.html
knowledge.broadcom.com